

Zhi-Hua Zhou

Machine Learning

Translated by Shaowu Liu

The slides of this chapter
are translated by Guo-Hua Wang



Springer

Chapter 13

Semi-Supervised Learning

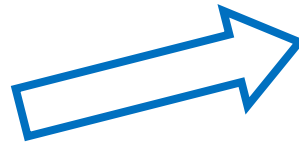
Outline

- Background
- Generative Methods
- Semi-Supervised SVM
- Graph-based Semi-Supervised Learning
- Disagreement-based Methods
- Semi-Supervised Clustering

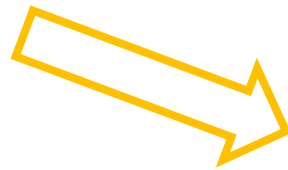
Background (Unlabeled Samples)



The melon farmer



eat



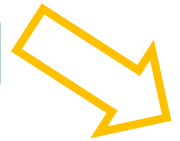
Background (Active Learning)



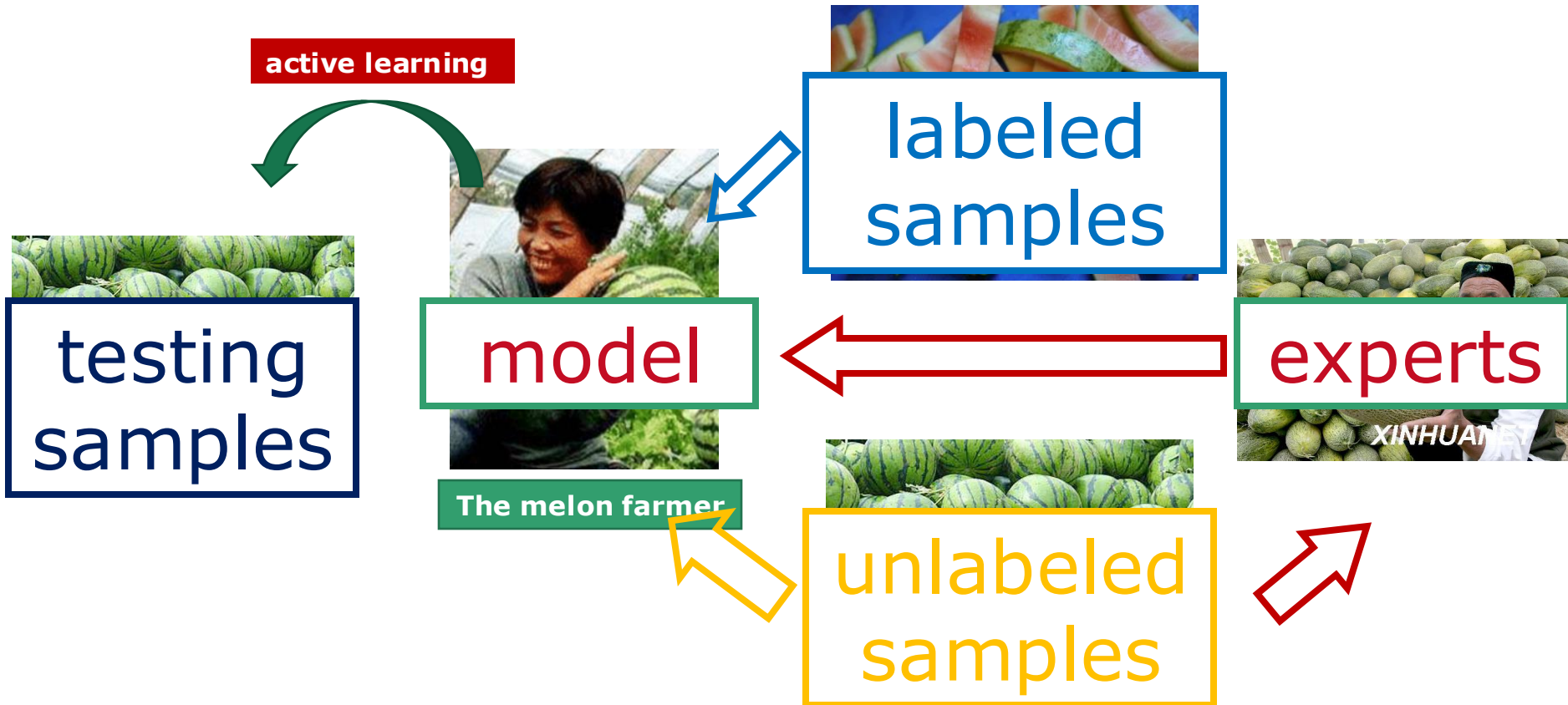
The melon farmer



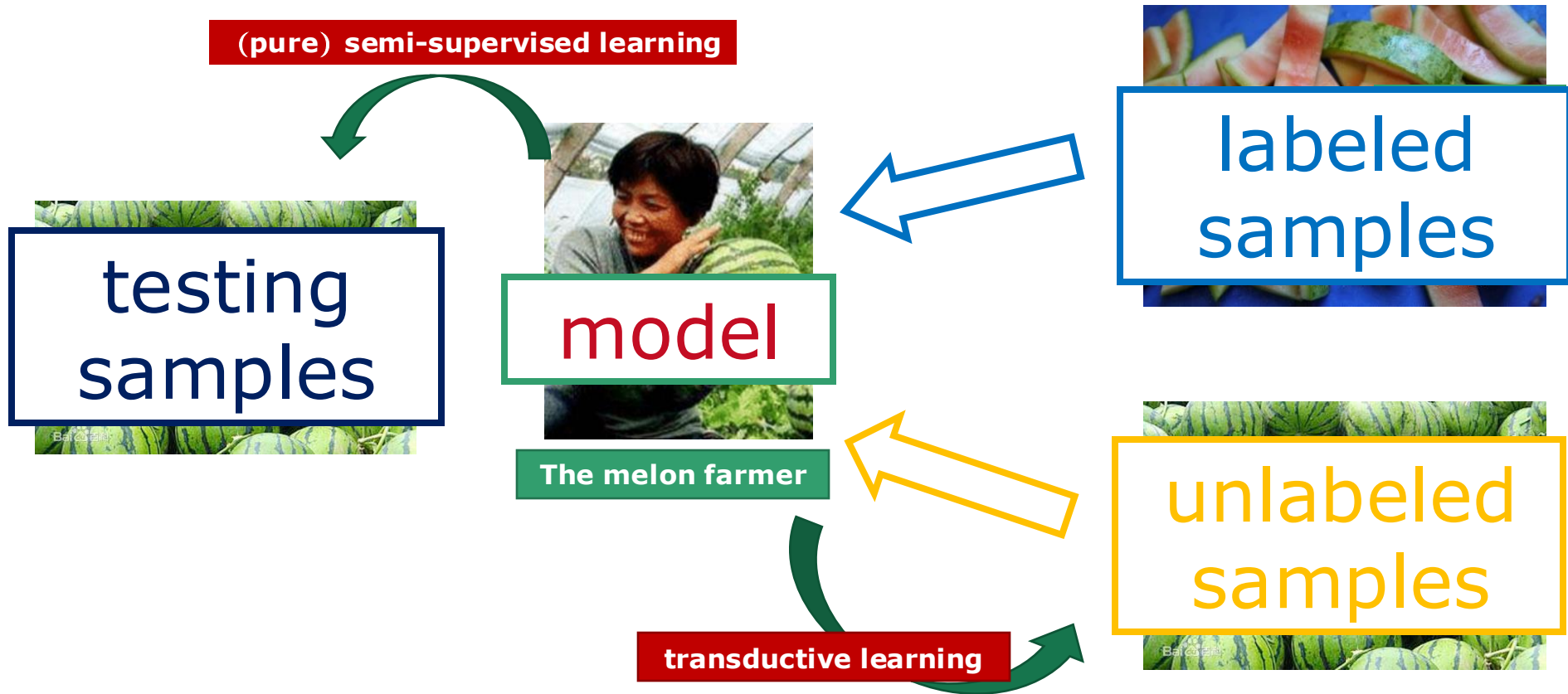
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Background (Active Learning)



Background (Semi-Supervised Learning)



The Effect of Unlabeled Samples

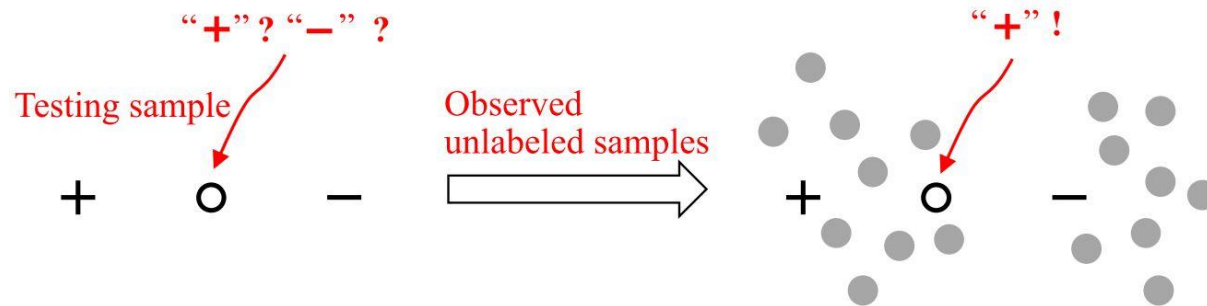


Fig. 13.1: An illustration of utilizing unlabeled samples. The gray dots on the right-hand side are unlabeled samples.

The Assumption of Unlabeled Samples

- ❑ To utilize unlabeled samples, we must make some assumptions to connect the underlying data distribution disclosed by the unlabeled samples to the labels. There are two common choices.
 - Cluster assumption:

Assume that the data contains clustering structures, and the samples within the same cluster belong to the same class.
 - Manifold assumption:

Assume that the samples are distributed on a manifold structure where neighboring samples have similar output values.

The manifold assumption can be seen as an extension of the clustering assumption

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Generative Methods

- Assume that the sample is generated by a Gaussian mixture model in which each Gaussian mixture component corresponds to a class:

$$p(\mathbf{x}) = \sum_{i=1}^k \alpha_i \cdot p(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

where, $\alpha_i \geq 0$, $\sum_{i=1}^k \alpha_i = 1$

- $p(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ is the probability of the sample $\mathbf{x}$ belonging to the i th Gaussian mixture component

$$p(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \frac{1}{(2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}_i|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)}$$

Generative Methods

□ Maximizing the posterior probability, we have:

$$f(\mathbf{x}) = \operatorname{argmax}_{j \in \mathcal{Y}} p(y = j | \mathbf{x})$$

$$= \operatorname{argmax}_{j \in \mathcal{Y}} \sum_{i=1}^k p(y = j, \Theta = i | \mathbf{x})$$

$$= \operatorname{argmax}_{j \in \mathcal{Y}} \sum_{i=1}^k p(y = j | \Theta = i, \mathbf{x}) \cdot p(\Theta = i | \mathbf{x})$$


$$p(y = j | \Theta = i)$$

We assume that each class corresponds to one Gaussian mixture component; the class of $\mathbf{x}$ only depends on the Gaussian mixture component $\mathbf{x}$ belongs to

Generative Methods

- Maximizing the posterior probability, we have:

$$f(\mathbf{x}) = \operatorname{argmax}_{j \in \mathcal{Y}} p(y = j | \mathbf{x})$$

$$= \operatorname{argmax}_{j \in \mathcal{Y}} \sum_{i=1}^k p(y = j, \Theta = i | \mathbf{x})$$

$$= \operatorname{argmax}_{j \in \mathcal{Y}} \sum_{i=1}^k p(y = j | \Theta = i, \mathbf{x}) \cdot p(\Theta = i | \mathbf{x})$$

where $p(\Theta = i | \mathbf{x}) = \frac{\alpha_i p(\mathbf{x} | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{i=1}^k \alpha_i p(\mathbf{x} | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}$



It does not involve the label: it can be expected to have a more accurate estimation with the help of a large amount of unlabeled samples

Generative Methods

- Assuming that all samples are i.i.d. and are generated by the same Gaussian mixture model. The log-likelihood of $D_l \cup D_u$ is formed as

$$\begin{aligned} \ln p(D_l \cup D_u) = & \sum_{(\mathbf{x}_j, y_j) \in D_l} \ln \left(\sum_{i=1}^k \alpha_i \cdot p(\mathbf{x}_j \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \cdot p(y_j \mid \Theta = i, \mathbf{x}_j) \right) \\ & + \sum_{\mathbf{x}_j \in D_u} \ln \left(\sum_{i=1}^k \alpha_i \cdot p(\mathbf{x}_j \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \right) . \end{aligned}$$

where $l \ll u$

Generative Methods

- The parameter estimation of Gaussian mixture model can be done using the EM algorithm with the following iterative update rules:
- **E-step:** compute the probabilities of an *unlabeled sample* belonging to *each Gaussian* mixture component based on the current model parameters

$$\gamma_{ji} = \frac{\alpha_i \cdot p(\mathbf{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{i=1}^N \alpha_i \cdot p(\mathbf{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}$$

Generative Methods

- **M-step:** update the model parameters based on γ_{ji}

number of labelled examples
belonging to the class i

$$\mu_i = \frac{1}{\sum_{\mathbf{x}_j \in D_u} \gamma_{ji} + l_i} \left(\sum_{\mathbf{x}_j \in D_u} \gamma_{ji} \mathbf{x}_j + \sum_{(\mathbf{x}_i, y_i) \in D_l \wedge y_i = i} \mathbf{x}_j \right)$$

$$\Sigma_i = \frac{1}{\sum_{\mathbf{x}_j \in D_u} \gamma_{ji} + l_i} \left(\sum_{\mathbf{x}_j \in D_u} \gamma_{ji} (\mathbf{x}_j - \mu_i)(\mathbf{x}_j - \mu_i)^T \right. \\ \left. + \sum_{(\mathbf{x}_i, y_i) \in D_l \wedge y_i = i} (\mathbf{x}_j - \mu_i)(\mathbf{x}_j - \mu_i)^T \right)$$

$$\alpha_i = \frac{1}{m} \left(\sum_{\mathbf{x}_j \in D_u} \gamma_{ji} + l_i \right)$$

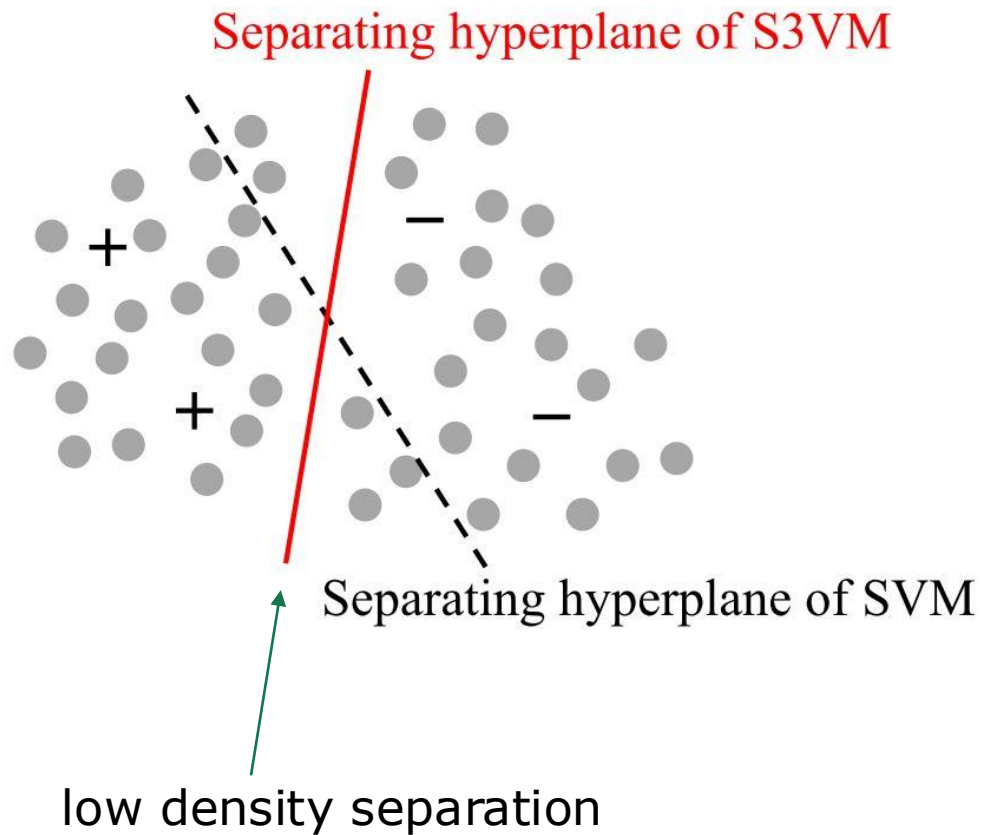
Generative Methods

- ❑ Other variants of generative methods can be derived by replacing the Gaussian mixture model by other models in the above process, such as **mixture of experts models** and **naïve Bayes models**.
- ❑ Generative methods are simple and easy to implement and can outperform other semi-supervised learning methods when **the labeled samples are extremely limited**.
- ❑ However, such methods heavily rely on **accurate model assumptions**, that is, the assumed generative model must match the ground-truth data distribution; otherwise, incorporating the unlabeled samples can even reduce the generalization performance.

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- Graph-based Semi-Supervised Learning
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Semi-Supervised SVM



Semi-Supervised SVM

- A well-known S3VM is Transductive Support Vector Machine (TSVM)

$$\min_{\mathbf{w}, b, \hat{\mathbf{y}}, \boldsymbol{\xi}} \quad \frac{1}{2} \|\mathbf{w}\|_2^2 + C_l \sum_{i=1}^l \xi_i + C_u \sum_{i=l+1}^m \xi_i \quad (\text{eq. 13.9})$$

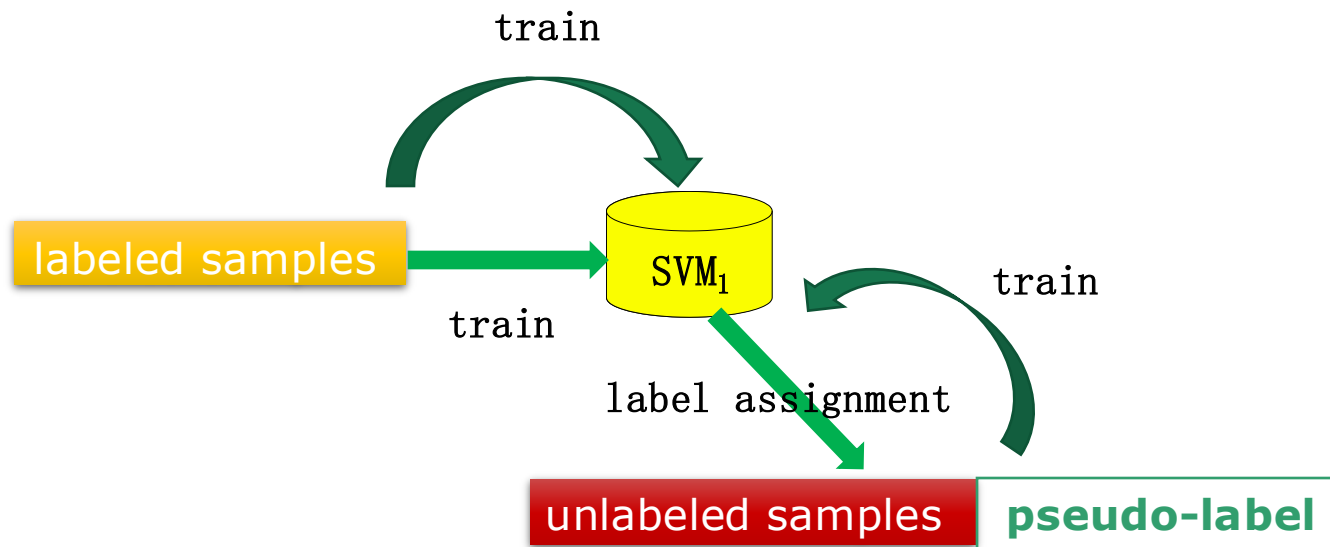
$$\text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = 1, \dots, l,$$

$$\hat{y}_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = l + 1, \dots, m,$$

$$\xi_i \geq 0, \quad i = 1, \dots, m,$$

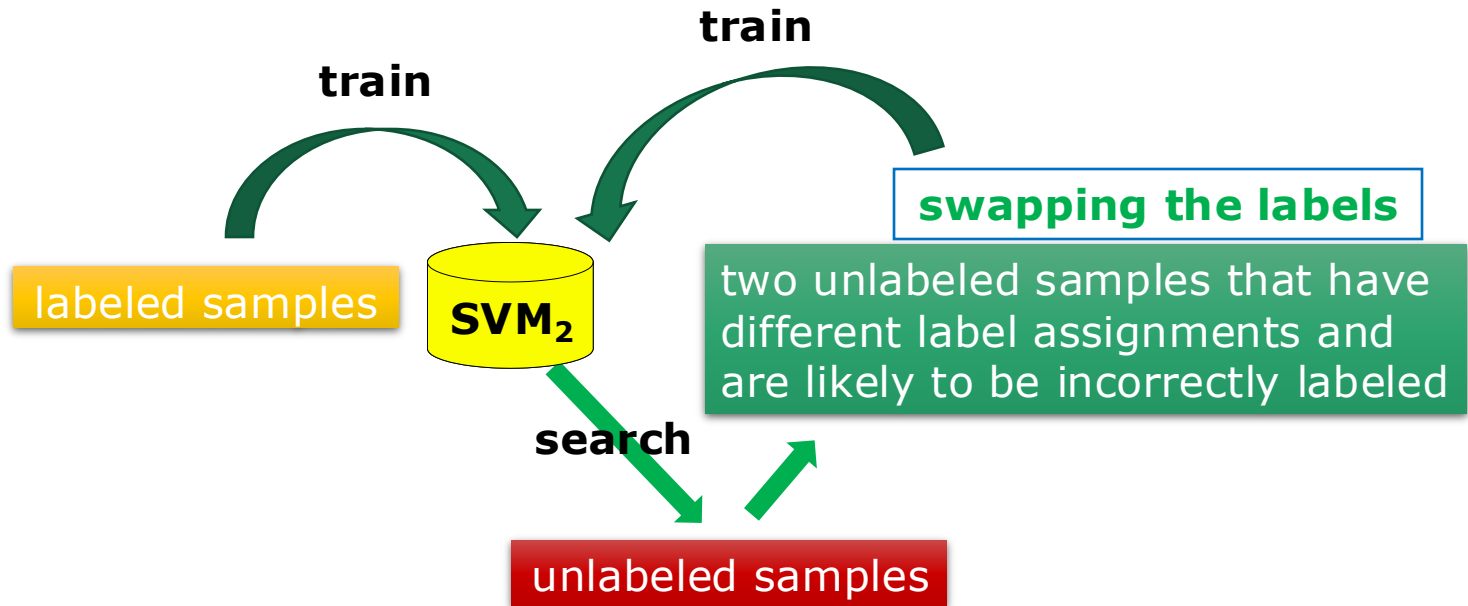
Semi-Supervised SVM

- TSVM takes an iterative local search strategy to find an approximate solution.



Semi-Supervised SVM

- TSVM takes an iterative local search strategy to find an approximate solution.



Semi-Supervised SVM

Algorithm 13.1 Transductive Support Vector Machine.

Input: Labeled data set $D_l = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_l, y_l)\}$;
 Unlabeled data set $D_u = \{\mathbf{x}_{l+1}, \mathbf{x}_{l+2}, \dots, \mathbf{x}_{l+u}\}$;
 Trade-off parameters C_l and C_u .

Process:

- 1: Train a SVM_l using D_l ;
- 2: Use SVM_l to predict the samples in D_u to get $\hat{\mathbf{y}} = (\hat{y}_{l+1}, \hat{y}_{l+2}, \dots, \hat{y}_{l+u})$;
- 3: Initialize $C_u \ll C_l$;
- 4: **while** $C_u < C_l$ **do**
- 5: Solve (13.9) using $D_l, D_u, \hat{\mathbf{y}}, C_l, C_u$ to get $(\mathbf{w}, b)$ and ξ ;
- 6: **while** $\exists \{i, j \mid (\hat{y}_i \hat{y}_j < 0) \wedge (\xi_i > 0) \wedge (\xi_j > 0) \wedge (\xi_i + \xi_j > 2)\}$ **do**
- 7: $\hat{y}_i = -\hat{y}_i$;
- 8: $\hat{y}_j = -\hat{y}_j$;
- 9: Solve (13.9) again using $D_l, D_u, \hat{\mathbf{y}}, C_l, C_u$ to get $(\mathbf{w}, b)$ and ξ ;
- 10: **end while**
- 11: $C_u = \min\{2C_u, C_l\}$. // increases the influence of unlabeled samples until $C_u = C_l$
- 12: **end while**

Output: Predictions of the unlabeled samples: $\hat{\mathbf{y}} = (\hat{y}_{l+1}, \hat{y}_{l+2}, \dots, \hat{y}_{l+u})$.

The pseudo-label of unlabeled samples might be incorrect: gradually increases the influence of unlabeled samples

$$\min_{\mathbf{w}, b, \hat{\mathbf{y}}, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 + C_l \sum_{i=1}^l \xi_i + C_u \sum_{i=l+1}^m \xi_i$$

$$\begin{aligned} \text{s.t. } & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = 1, \dots, l, \\ & \hat{y}_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = l+1, \dots, m, \\ & \xi_i \geq 0, \quad i = 1, \dots, m, \end{aligned}$$

finds two unlabeled samples that have different label assignments and are likely to be incorrectly labeled

Semi-Supervised SVM

- ❑ The learning process of TSVM may suffer from **the class imbalance problem**, in which the number of samples in one class is far more than the number of samples in the other class.
- ❑ To alleviate this problem, we can slightly adjust Algorithm. Specifically, we split the term C_u in the optimization objective to C_u^+ and C_u^- . TSVM is initialized with

$$C_u^+ = \frac{u_-}{u_+} C_u^-$$

number of unlabeled samples with negative pseudo-labels

number of unlabeled samples with positive pseudo-labels

Semi-Supervised SVM

- ❑ The iterative updating process is an **expensive large-scale** optimization problem because the algorithm iterates over each pair of unlabeled samples that are likely to be incorrectly labeled.
- ❑ Hence, one research focus of S3VM is **the design of efficient optimization strategies**.
- ❑ Such as LDS [Chapelle and Zien, 2005], which uses gradient descent on graph kernel functions, and meanS3VM [Li et al., 2009], which uses label mean estimation.

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Graph-based Semi-Supervised Learning

- ❑ Training data sets can often be encoded into a graph structure, in which each node corresponds to a sample, and an edge connects two nodes if their corresponding samples are **highly similar** (or correlated). The **strength** of an edge indicates **the degree of similarity** (or correlation).
- ❑ Imagine that the nodes of labeled samples are colored, and the nodes of unlabeled samples are uncolored. Then, semi-supervised learning is like the process of spreading the *color* over the graph.
- ❑ Since a graph corresponds to a matrix, we can derive and analyze semi-supervised learning algorithms via matrix operations.

Graph-based Semi-Supervised Learning

- We start by constructing a graph $G = (V, E)$ from $D_l \cup D_u$, where

$$V = \{x_1, \dots, x_l, x_{l+1}, \dots, x_{l+u}\}$$

- the edge set E can be represented as an affinity matrix which is often defined based on the Gaussian function:

$$W_{ij} = \begin{cases} \exp\left(\frac{-\|x_i - x_j\|_2^2}{2\sigma^2}\right), & \text{if } i \neq j ; \\ 0 & , \text{ otherwise ,} \end{cases}$$

Graph-based Semi-Supervised Learning

- From graph $G = (V, E)$, we aim to learn a real-valued function $f : V \rightarrow \mathbb{R}$
- Intuitively, similar samples share similar labels, and hence we can define an *energy function* [Zhu et al., 2003]:

$$\begin{aligned} E(f) &= \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \mathbf{W}_{ij} (f(\mathbf{x}_i) - f(\mathbf{x}_j))^2 \\ &= \frac{1}{2} \left(\sum_{i=1}^m d_i f^2(\mathbf{x}_i) + \sum_{j=1}^m d_j f^2(\mathbf{x}_j) - 2 \sum_{i=1}^m \sum_{j=1}^m \mathbf{W}_{ij} f(\mathbf{x}_i) f(\mathbf{x}_j) \right) \\ &= \mathbf{f}^T (\mathbf{D} - \mathbf{W}) \mathbf{f} \end{aligned}$$

Graph-based Semi-Supervised Learning

- We partition the matrix and group the elements into four blocks:

$$\begin{aligned} E(f) &= (f_l^\top \ f_u^\top) \left(\begin{bmatrix} \mathbf{D}_{ll} & \mathbf{0}_{lu} \\ \mathbf{0}_{ul} & \mathbf{D}_{uu} \end{bmatrix} - \begin{bmatrix} \mathbf{W}_{ll} & \mathbf{W}_{lu} \\ \mathbf{W}_{ul} & \mathbf{W}_{uu} \end{bmatrix} \right) \begin{bmatrix} f_l \\ f_u \end{bmatrix} \\ &= f_l^\top (\mathbf{D}_{ll} - \mathbf{W}_{ll}) f_l - 2f_u^\top \mathbf{W}_{ul} f_l + f_u^\top (\mathbf{D}_{uu} - \mathbf{W}_{uu}) f_u . \end{aligned}$$

- By setting $\frac{\partial E(f)}{\partial f_u} = 0$, we have

$$f_u = (\mathbf{D}_{uu} - \mathbf{W}_{uu})^{-1} \mathbf{W}_{ul} f_l$$

Graph-based Semi-Supervised Learning

$$\mathbf{P} = \mathbf{D}^{-1}\mathbf{W} = \begin{bmatrix} \mathbf{D}_{ll}^{-1} & \mathbf{0}_{lu} \\ \mathbf{0}_{ul} & \mathbf{D}_{uu}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{W}_{ll} & \mathbf{W}_{lu} \\ \mathbf{W}_{ul} & \mathbf{W}_{uu} \end{bmatrix} = \begin{bmatrix} \mathbf{D}_{ll}^{-1}\mathbf{W}_{ll} & \mathbf{D}_{ll}^{-1}\mathbf{W}_{lu} \\ \mathbf{D}_{uu}^{-1}\mathbf{W}_{ul} & \mathbf{D}_{uu}^{-1}\mathbf{W}_{uu} \end{bmatrix}$$

$$\mathbf{P}_{uu} = \mathbf{D}_{uu}^{-1}\mathbf{W}_{uu}, \mathbf{P}_{ul} = \mathbf{D}_{uu}^{-1}\mathbf{W}_{ul}$$

$$\mathbf{f}_u = (\mathbf{D}_{uu}(\mathbf{I} - \mathbf{D}_{uu}^{-1}\mathbf{W}_{uu}))^{-1}\mathbf{W}_{ul}\mathbf{f}_l$$

$$= (\mathbf{I} - \mathbf{D}_{uu}^{-1}\mathbf{W}_{uu})^{-1}\mathbf{D}_{uu}^{-1}\mathbf{W}_{ul}\mathbf{f}_l$$

$$= (\mathbf{I} - \mathbf{P}_{uu})^{-1}\mathbf{P}_{ul}\mathbf{f}_l$$

Graph-based Semi-Supervised Learning

- The above process is a *label propagation* method for binary classification. Next, we introduce another label propagation method [Zhou et al., 2004] that is applicable to multiclass classification.
- We start by constructing a graph $G = (V, E)$ from $D_l \cup D_u$, where $V = \{\mathbf{x}_1, \dots, \mathbf{x}_l, \mathbf{x}_{l+1}, \dots, \mathbf{x}_{l+u}\}$ is the node set and E is the edge set.
- We define a $(l+u) \times |\mathcal{Y}|$ non-negative label matrix $\mathbf{F} = (\mathbf{F}_1^\top, \dots, \mathbf{F}_{|\mathcal{Y}|}^\top)^\top$, where the i -th row $\mathbf{F}_i = (F_{i1}, \dots, F_{i|\mathcal{Y}|})$ is the label vector of sample $\mathbf{x}_i$, and the classification rule is:

$$y_i = \operatorname{argmax}_{1 \leq j \leq |\mathcal{Y}|} F_{ij}$$

- We initialize $\mathbf{F}$ as

$$\mathbf{F}(0) = \mathbf{Y}_{ij} = \begin{cases} 1, & \text{if } (1 \leq i \leq l) \wedge (y_i = j) ; \\ 0, & \text{otherwise ,} \end{cases}$$

Graph-based Semi-Supervised Learning

- We construct a label propagation matrix $S = D^{-\frac{1}{2}}WD^{-\frac{1}{2}}$ based on W , where $D^{-\frac{1}{2}} = \left(\frac{1}{\sqrt{d_1}}, \dots, \frac{1}{\sqrt{d_{l+u}}} \right)$. Then, we have the iterative formula:

$$\mathbf{F}(t+1) = \alpha \mathbf{S} \mathbf{F}(t) + (1 - \alpha) \mathbf{Y}$$

- Iterating until convergence, we have:

$$\mathbf{F}^* = \lim_{t \rightarrow \infty} \mathbf{F}(t) = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{S})^{-1} \mathbf{Y}$$

Graph-based Semi-Supervised Learning

- Note that the algorithm corresponds to the following regularization framework:

$$\min_{\mathbf{F}} \frac{1}{2} \left(\sum_{i,j=1}^{l+u} \mathbf{W}_{ij} \left\| \frac{1}{\sqrt{d_i}} \mathbf{F}_i - \frac{1}{\sqrt{d_j}} \mathbf{F}_j \right\|^2 \right) + \mu \sum_{i=1}^l \|\mathbf{F}_i - \mathbf{Y}_i\|^2$$

- When $\mu = \frac{1-\alpha}{\alpha}$, the optimal solution equals to the converged solution $\mathbf{F}^*$.

Graph-based Semi-Supervised Learning

Algorithm 13.2 Iterative Label Propagation.

Input: Labeled data set $D_l = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_l, y_l)\}$;
Unlabeled data set $D_u = \{\mathbf{x}_{l+1}, \mathbf{x}_{l+2}, \dots, \mathbf{x}_{l+u}\}$;
Graph construction parameter σ ;
Trade-off parameter α .

Process:

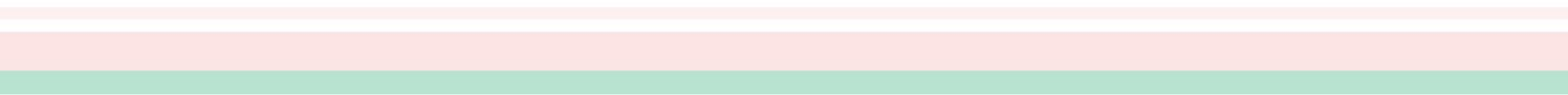
- 1: Use (13.11) and parameter σ to obtain $\mathbf{W}$;
- 2: Use $\mathbf{W}$ to construct the label propagation matrix $\mathbf{S} = \mathbf{D}^{-\frac{1}{2}} \mathbf{W} \mathbf{D}^{-\frac{1}{2}}$;
- 3: Initialize $\mathbf{F}(0)$ according to (13.18);
- 4: $t = 0$;
- 5: **repeat**
- 6: $\mathbf{F}(t+1) = \alpha \mathbf{S} \mathbf{F}(t) + (1 + \alpha) \mathbf{Y}$;
- 7: $t = t + 1$;
- 8: **until** Converged to $\mathbf{F}^*$
- 9: **for** $i = l + 1, l + 2, \dots, l + u$ **do**
- 10: $\hat{y}_i = \arg \max_{1 \leq j \leq |\mathcal{Y}|} (\mathbf{F}^*)_{ij}$.
- 11: **end for**

Output: Predictions of the unlabeled samples: $\hat{\mathbf{y}} = (\hat{y}_{l+1}, \hat{y}_{l+2}, \dots, \hat{y}_{l+u})$.

Graph-based Semi-Supervised Learning

- ❑ The concepts in graph-based semi-supervised learning methods are well-defined, and the algorithms' properties can be easily analyzed from the matrix operations.
- ❑ However, such methods also have some deficiencies. Firstly, the **memory cost can be very high**.
- ❑ The graphs are constructed from the **training samples**, making it difficult to identify the positions of new samples. Hence, with new samples arrived, we need to re-construct the graph and redo the label propagation, or introduce an extra prediction mechanism.

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- 

Disagreement-based Methods

- ❑ *Disagreement-based methods* train multiple learners and the *disagreement* between learners is crucial for the exploitation of unlabeled data.
- ❑ Co-training [Blum and Mitchell, 1998] is a representative disagreement-based methods. Since it was originally designed for *multi-view data*, it is also a representative method for *multi-view learning*.

Disagreement-based Methods

Image View

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[\[Interest\]](#) [\[Career\]](#) [\[Education\]](#) [\[Award\]](#) [\[Activity\]](#) [\[Publication\]](#) [\[Course\]](#) [\[Student and Postdoc\]](#) [\[LAMDA Group\]](#)

Text View

Research Interest

I have wide research interests, mainly including *artificial intelligence, machine learning, data mining, pattern recognition, evolutionary computation and multimedia retrieval*, among which **machine learning and data mining** problem of how to enable computing machines to handle "ambiguity".

Currently I am interested in the following ML/DM topics:

- [Multi-label learning](#)
- [Multi-instance learning](#)
- [Semi-supervised and active learning](#)
- [Cost-sensitive and class-imbalance learning](#)
- [Metric learning, dimensionality reduction and feature selection](#)
- [Ensemble learning](#)
- [Structure learning and clustering](#)

For applications, I am mainly interested in the following areas:

- [Image retrieval](#)

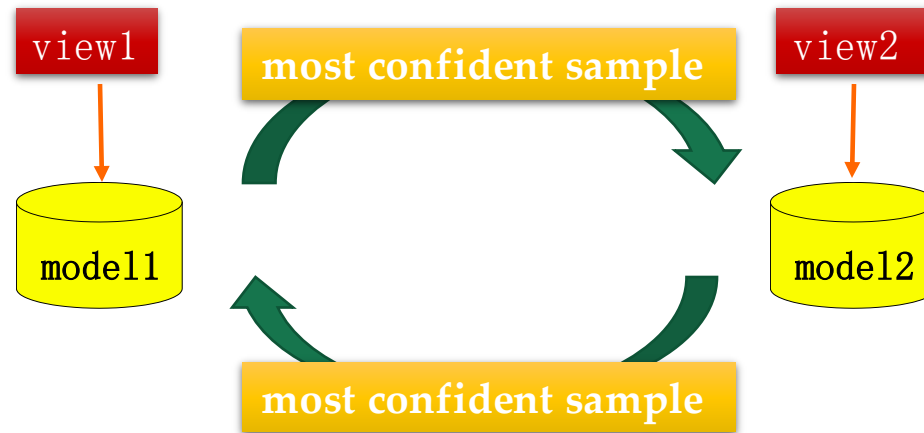


Z.-H. Zhou, *Ensemble Methods: Foundations and Algorithms*, Boca Raton, FL: Chapman & Hall/CRC, 2012. (ISBN 978-1-439-83003-1)

Two views in the website classification task

Disagreement-based Methods

- Co-training can exploit the *compatibility and complementarity* of multi-view data. Suppose that the data has two *sufficient and conditionally independent* views.



Disagreement-based Methods

The superscript of $\mathbf{x}_i$ indicates the two views rather than ordinal relationships, that is, $\langle \mathbf{x}_1^1 \mathbf{x}_1^2 \rangle$ and $\langle \mathbf{x}_1^2 \mathbf{x}_1^1 \rangle$ refer to the same sample.

Letting $p, n \ll s$.

Algorithm 13.3 Co-training.

Input: Labeled data set $D_l = \{(\langle \mathbf{x}_1^1 \mathbf{x}_1^2 \rangle, y_1), \dots, (\langle \mathbf{x}_l^1 \mathbf{x}_l^2 \rangle, y_l)\}$;
 Unlabeled data set $D_u = \{(\langle \mathbf{x}_{l+1}^1 \mathbf{x}_{l+2}^2 \rangle, \dots, \langle \mathbf{x}_{l+u}^1 \mathbf{x}_{l+u}^2 \rangle)\}$;
 Buffer pool size s ;
 Number of positive samples in each round p ;
 Number of negative samples in each round n ;
 Base learning algorithm $\mathcal{L}$;
 Number of learning rounds T .

Procedure

Train h_j on the view j using the labeled samples.	<pre> 7: for $j = 1, 2$ do 8: $h_j \leftarrow \mathcal{L}(D_l^j)$; 9: Check the classification confidences of h_j on $D_s^j = \{\langle \mathbf{x}_i^j, \mathbf{x}_i^{3-j} \rangle \in D_s\}$, and select p samples $D_p \subset D_s$ with the highest confidence of being positive and n samples $D_n \subset D_s$ with the highest confidence of being negative; 10: Use D_p^j to generate pseudo-labeled positive samples $\tilde{D}_p^{3-j} = \{(\langle \mathbf{x}_i^{3-j}, +1 \rangle \mid \langle \mathbf{x}_i^j \in D_p^j\}$; 11: Use D_n^j to generate pseudo-labeled negative samples $\tilde{D}_n^{3-j} = \{(\langle \mathbf{x}_i^{3-j}, -1 \rangle \mid \langle \mathbf{x}_i^j \in D_n^j\}$; 12: $D_s = D_s \setminus (D_p \cup D_n)$; 13: end for </pre>
Extend the labeled data set.	<pre> 12: $D_s = D_s \setminus (D_p \cup D_n)$; 13: end for 17: for $j = 1, 2$ do 18: $D_l^j = D_l^j \cup (\tilde{D}_p^j \cup \tilde{D}_n^j)$; 19: end for </pre>
	<pre> 19: end for 20: Randomly select $2p + 2n$ samples from D_u and add them to D_s. 21: end if 22: end for Output: Classifiers h_1 and h_2. </pre>

Disagreement-based Methods

- ❑ Though co-training looks simple, theoretical studies [Blum and Mitchell, 1998] showed that the generalization performance of weak classifiers can be boosted to arbitrary high using unlabeled samples given that the views are **sufficient and conditionally independent**.
- ❑ In practice, however, since the views rarely hold conditional independence. The improvement in performance is often not that significant as in theory. Notably, the performance of weak learners can still be considerably improved even with weaker conditions [Zhou and Li, 2010].

Disagreement-based Methods

- ❑ Co-training was originally designed for multi-view data, but some variants have been developed for single-view data.
- ❑ In these variants, different learners are realized by **different algorithms** [Goldman and Zhou, 2000], **different data sampling** [Zhou and Li, 2005b], or even **different parameter settings** [Zhou and Li, 2005a]. The performance can still be improved in these variants by utilizing unlabeled samples.
- ❑ Later on, theoretical studies revealed that multi-view data is not necessary for such algorithms; instead, as long as **the weak learners have significant disagreement (or difference)**, exchanging pseudo-labels can improve generalization performance [Zhou and Li, 2010].

Disagreement-based Methods

- ❑ By choosing the suitable base learners, disagreement-based methods are less affected by model assumptions, non-convexity of loss function, and data scale problems. These simple and effective methods have a solid theoretical foundation and a wide scope of applications.
- ❑ To use them, we need to train multiple learners with significant differences and acceptable performance, but when there are very limited labeled samples, especially when the data does not have multiple views, it is not easy to do without ingenious design.

Outline

- Background
- Generative Methods
- Semi-Supervised SVM
- Graph-based Semi-Supervised Learning
- Disagreement-based Methods
- Semi-Supervised Clustering

Semi-Supervised Clustering

- ❑ Clustering is a typical unsupervised learning problem. In practice, however, we often have access to some additional supervision information, which can be utilized to obtain better clustering results via *semi-supervised clustering*.
- ❑ Typically, there are two kinds of supervision information in clustering tasks:
 - The first kind is *must-link* and *cannot-link* constraints, where must-link means the samples must belong to the same cluster, and cannot-link means the samples must not belong to the same cluster.
 - The second kind of supervision information refers to a small number of *labeled samples*.

Semi-Supervised Clustering

- ❑ Constrained k-means [Wagstaff et al., 2001] is a representative semi-supervised clustering algorithm that utilizes supervision information in the form of must-link and cannot-link constraints.
- ❑ Constrained k-means, which is an extension of k-means, ensures that the constraints in *must-link set* and *cannot-link set* are satisfied during the clustering process; otherwise it will return an error when the constraints are violated.

Semi-Supervised Clustering

Algorithm 13.4 Constrained k -means Clustering.

Input: Data set $D = \{x_1, x_2, \dots, x_n\}$.

```

8:   while  $\neg$ is_merged do
9:       Use  $\mathcal{K}$  to find the nearest cluster of  $x_i$ :  $r = \arg \min_{j \in \mathcal{K}} d_{ij}$ ;
10:      Check if putting  $x_i$  into  $C_r$  violates any constraint in  $\mathcal{M}$  or  $\mathcal{C}$ ;
11:      if  $\neg$ is_violated then
12:           $C_r = C_r \cup \{x_i\}$ ;
13:          is_merged=true
14:      else
15:           $\mathcal{K} = \mathcal{K} \setminus \{r\}$ ;
16:          if  $\mathcal{K} = \emptyset$  then
17:              break and return an error message
18:          end if
19:      end if
20:   end while

```

Not violated, choose the nearest cluster

Violated, try the secondary nearest cluster

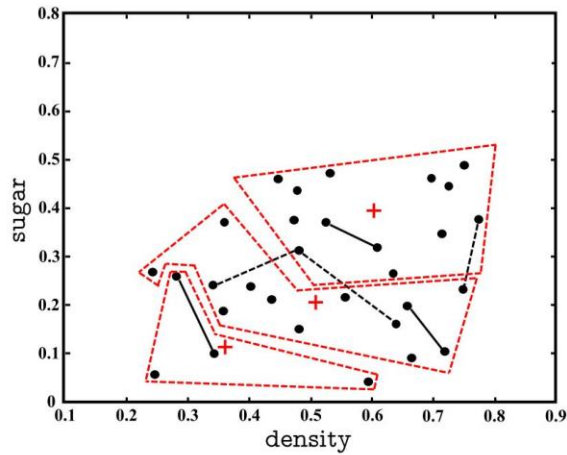
23: $\mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} x$

24: end for

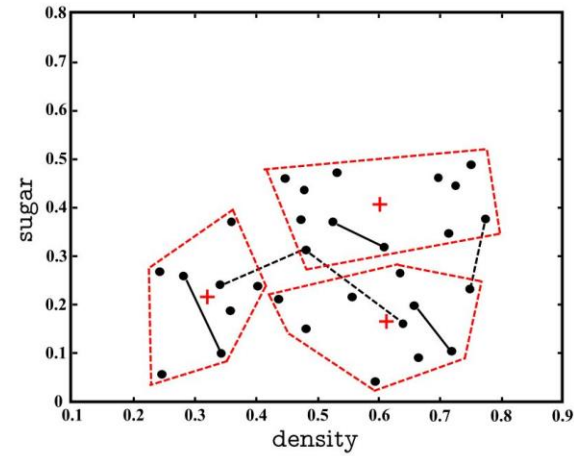
25: until All mean vectors remain unchanged

Output: Clusters $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$

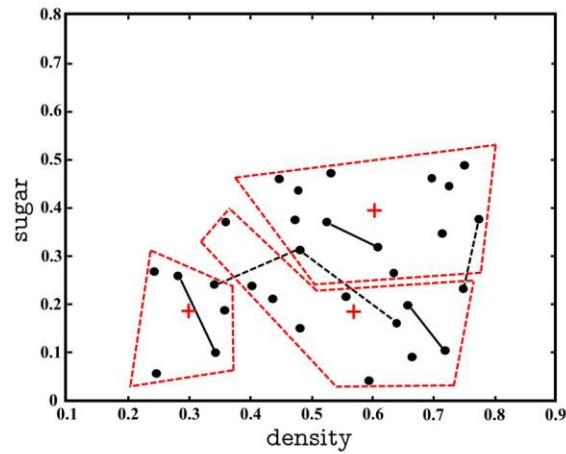
Semi-Supervised Clustering



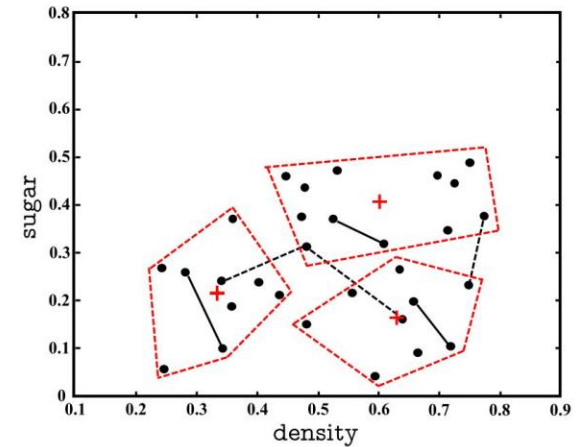
(a) After 1 iteration.



(c) After 3 iterations.



(b) After 2 iterations.



(d) After 4 iterations.

Semi-Supervised Clustering

- Supervision information also exists in the form of a small number of labeled samples.
- It is straightforward to use such supervision information: we can use the labeled samples as the *seeds* to initialize the k centroids of the k -means algorithm, and these seeds are retained in their clusters during the learning process. Such a procedure gives the Constrained Seed k -means [Basu et al., 2002].

Semi-Supervised Clustering

Algorithm 13.5 Constrained Seed k -means Clustering.

Input: Data set $D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$;

Initialize the cluster centroids using labeled samples.

```
1: for  $j = 1, 2, \dots, k$  do  
2:    $\mu_j = \frac{1}{|S_j|} \sum_{\mathbf{x} \in S_j} \mathbf{x}$ ;  
3: end for
```

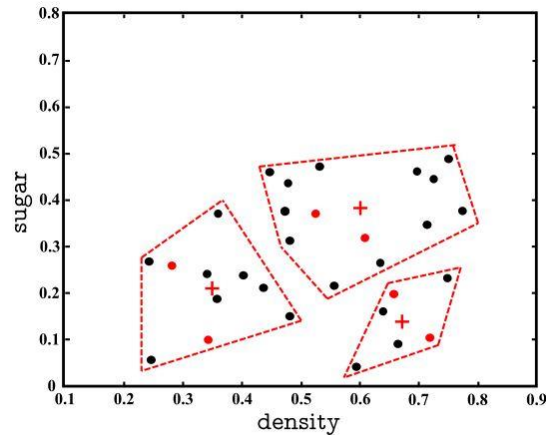
Initialize the k clusters using labeled samples.

```
6:   for  $j = 1, 2, \dots, k$  do  
7:     for all  $\mathbf{x} \in S_j$  do  
8:        $C_j = C_j \cup \{\mathbf{x}\}$ ;  
9:     end for
```

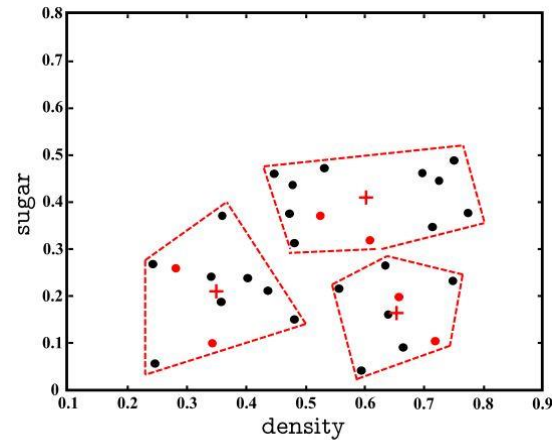
Update the mean vectors.

```
    $d_{ij} = \|\mathbf{x}_i - \mu_j\|_2$ ;  
13:   Find the nearest cluster of  $\mathbf{x}_i$ :  $r = \arg \min_{j \in \{1, 2, \dots, k\}} d_{ij}$ ;  
14:   Move  $\mathbf{x}_i$  to the corresponding cluster:  $C_r = C_r \cup \{\mathbf{x}_i\}$ ;  
15: end for  
16: for  $j = 1, 2, \dots, k$  do  
17:    $\mu_j = \frac{1}{|C_j|} \sum_{\mathbf{x} \in C_j} \mathbf{x}$ ;  
18: end for  
19: until All mean vectors remain unchanged  
Output: Clusters  $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$ .
```

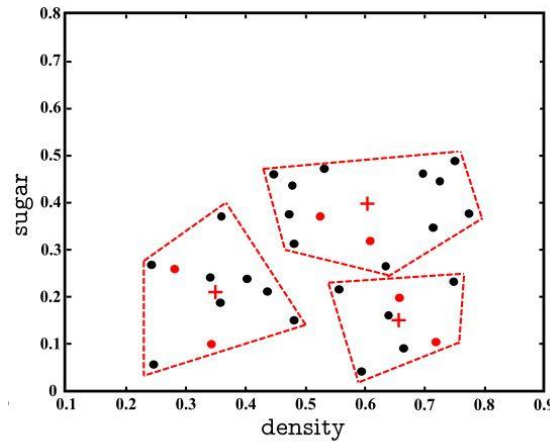
Semi-Supervised Clustering



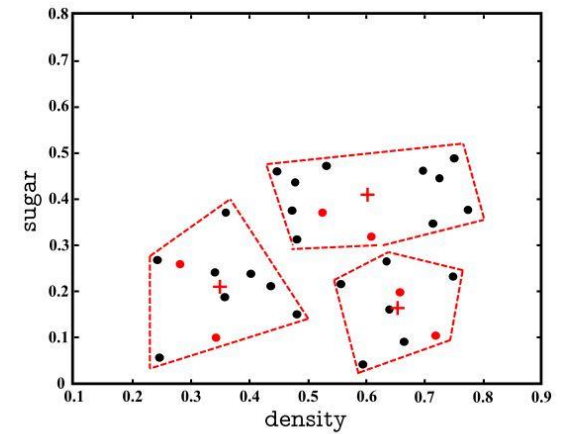
(a) After 1 iteration.



(c) After 3 iterations.



(b) After 2 iterations.



(d) After 4 iterations.

Further Reading

- ❑ Utilizing unlabeled samples through semi-supervised learning does **not necessarily guarantee the improvement in generalization performance**, and it may even decrease the performance sometimes. For generative methods, it is believed that **inappropriate model assumptions** cause performance degeneration [Cozman and Cohen, 2002], and hence such methods rely on sufficiently reliable domain knowledge.

Further Reading

- ❑ For semi-supervised SVM, the performance degeneration is believed to be caused by the **existence of multiple “low-density separations”** in the training set such that the learning algorithm may make a poor selection. S4VM [Li and Zhou, 2015] improved the “safeness” of semi-supervised SVM by optimizing the worst-case performance by exploiting the multiple low-density separators. A general approach to “safe” semi-supervised learning has yet to be developed.

Further Reading

- Besides classification and clustering problems, semi-supervised learning has also been widely used in other machine learning problems, such as **semi-supervised regression** [Zhou and Li, 2005a] and **dimensionality reduction** [Zhang et al., 2007]. See [Chapelle et al., 2006; Zhu, 2006] for more information about semi-supervised learning. Detailed discussions on disagreement-based methods can be found in [Zhou and Li, 2010]. [Settles, 2009] provided an introduction to active learning.

Thanks!
